

CM515_Assignment_13_mcewen

Gavin McEwen

Instructions

The aim of this assignment is to provide you with an opportunity to sharpen your skills in using `ggplot()`. While some of the plots you create may resemble those we covered in class, it is essential that they are original.

Rules for the Assignment:

1. All data sets except iris are fair game. That includes past data sets used in the class, sets built into R, your own data, and even online data.
2. All graphs must include axis labels, plot title, a theme of your choice, and a brief description/interpretation of the plot (2-3 sentences).

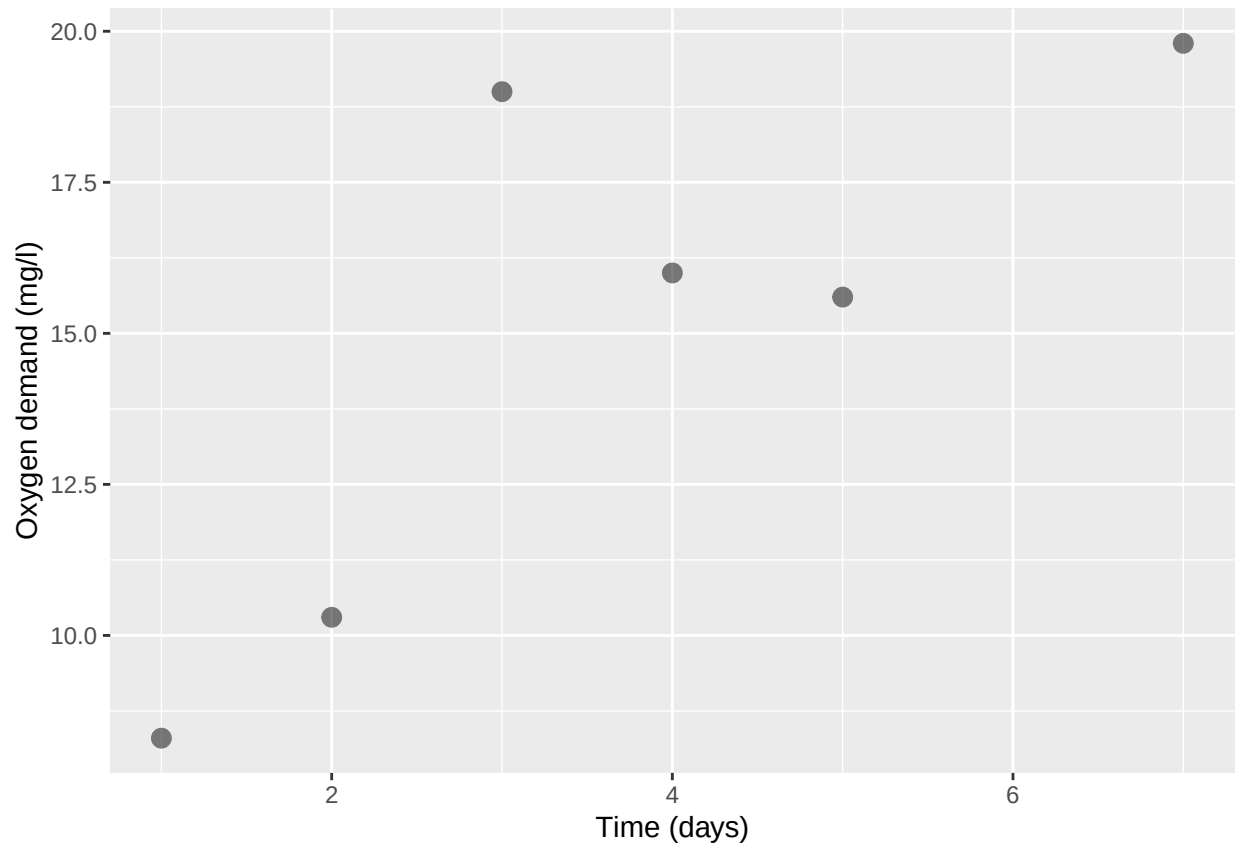
It's worth noting that `ggplot()` is a widely-used tool, and there are numerous online resources available for you to explore. We strongly recommend that you take advantage of these resources to deepen your understanding of `ggplot()`.

Knit the document into a PDF and submit it to Canvas by 04/27/2026 at 11:59 pm.

Load Packages and Data

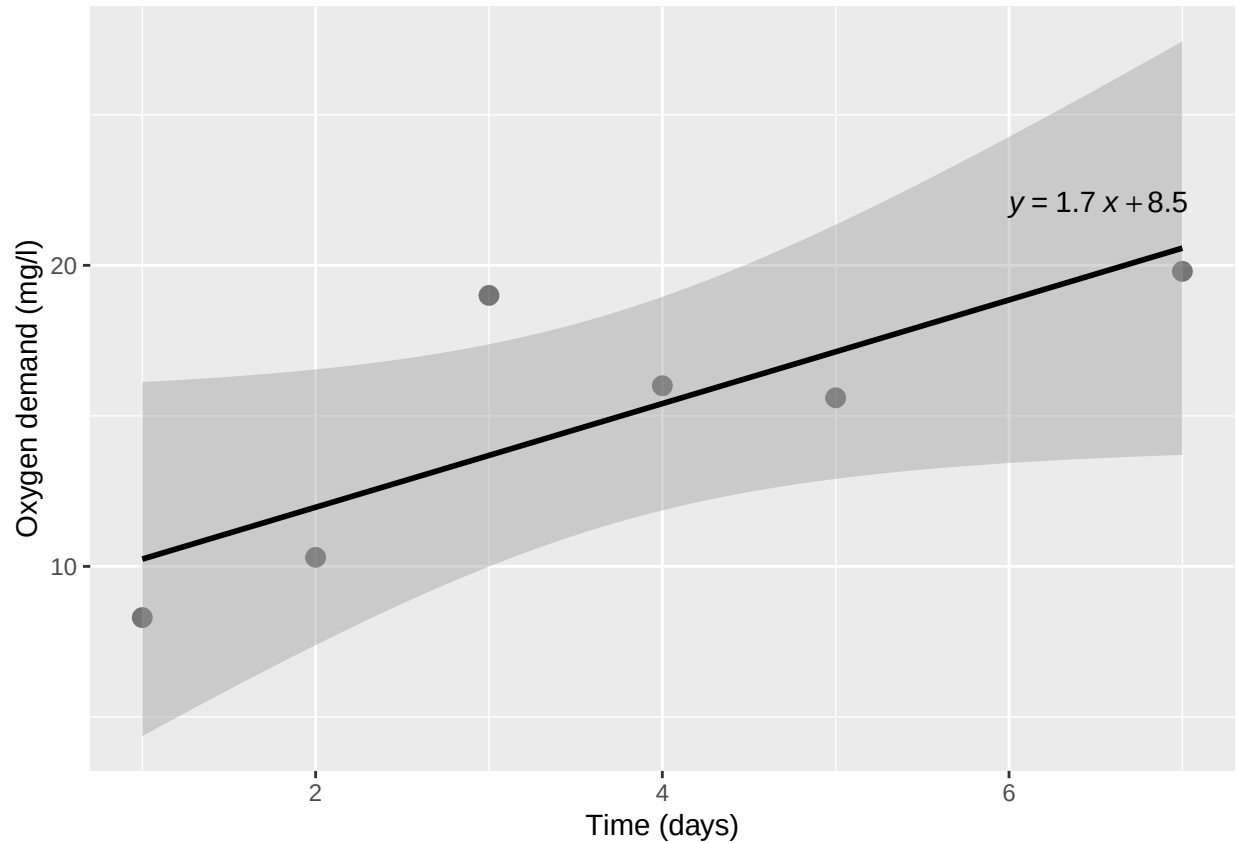
1. Make a Scatter Plot With Customized Point Size and Transparency (3pts)

Description and Interpretation: I loaded a dataset containing biochemical oxygen demand versus time into a scatter plot. As time increases, biochemical oxygen demand also increases.



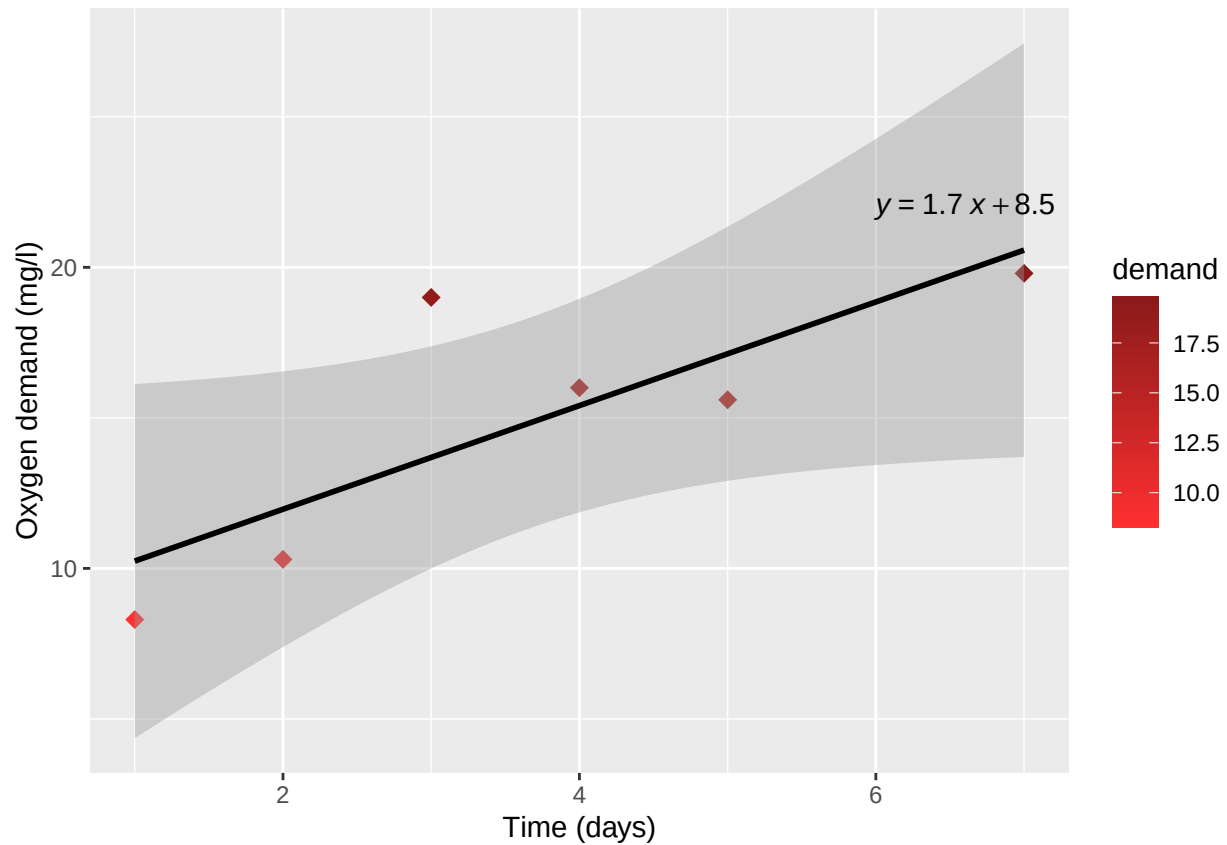
2. Fit a Line Through Your Scatter Plot From 1. (3pts)

Description and Interpretation: I fit a linear line through the 6 data points. The best-fit line has a slope of 1.7 (mg/l)/day and a y-intercept of 8.5 mg/l.



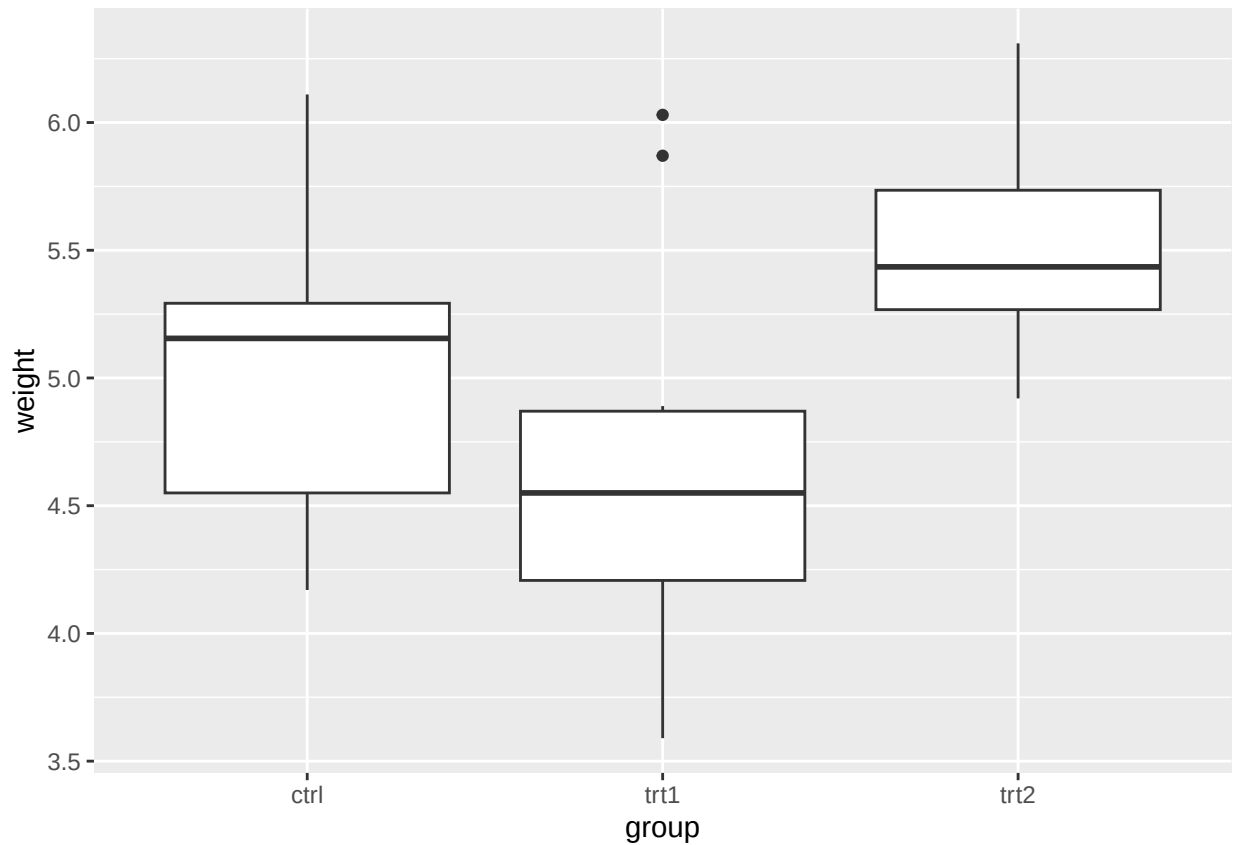
3. Set the transparency, shape, and color of your graph from 1 based on expressions. Scale the color. (3pts)

Description and Interpretation: I set the shape of each point on my plot to diamonds. I also decreased the transparency (i.e., increased the opacity) and scales the colors along a “firebrick” gradient, which offered good contrast.



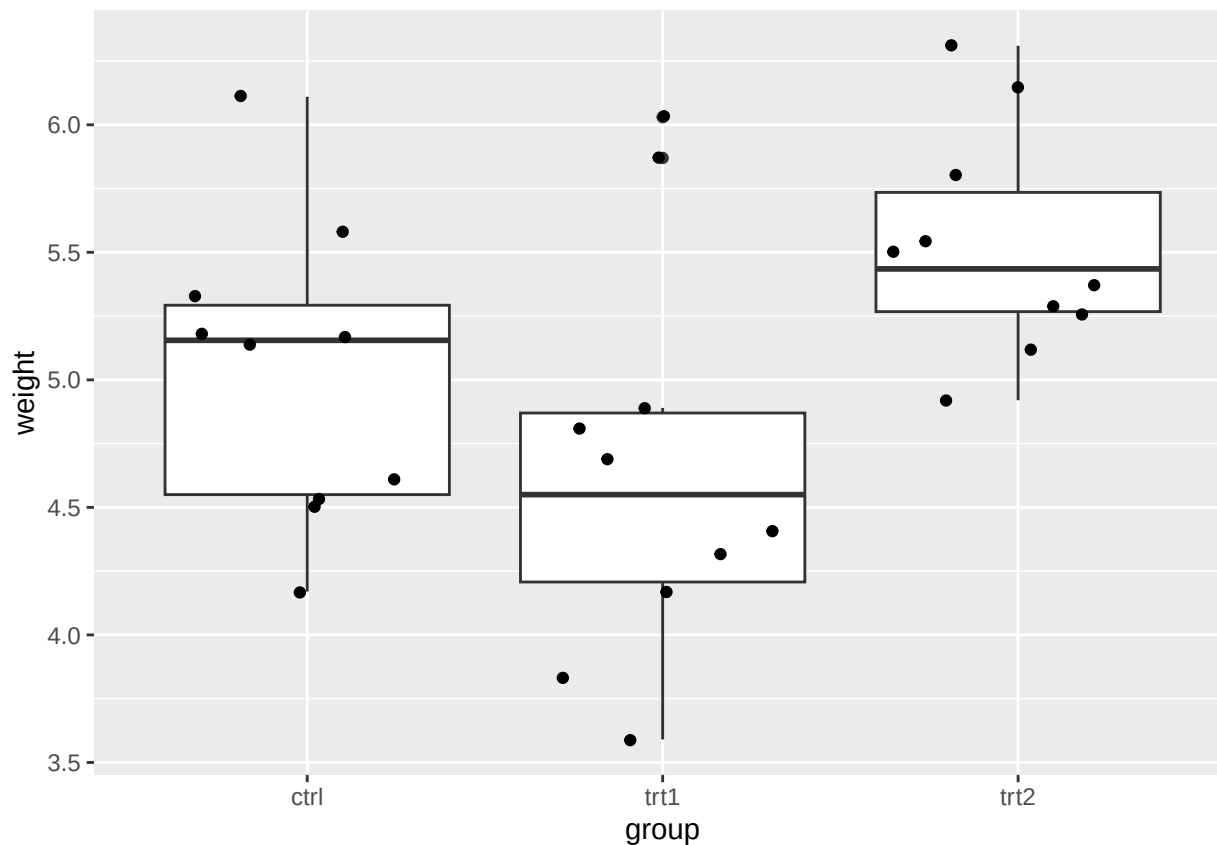
4. Make a Boxplot With Customized Boxplot Width (3pts)

Description and Interpretation: I loaded a dataset containing dry weight yields of plants under different treatment conditions. The units of weight are unspecified. Treatment 2 increased dry weight versus control, while treatment 1 decreased try weight versus control.



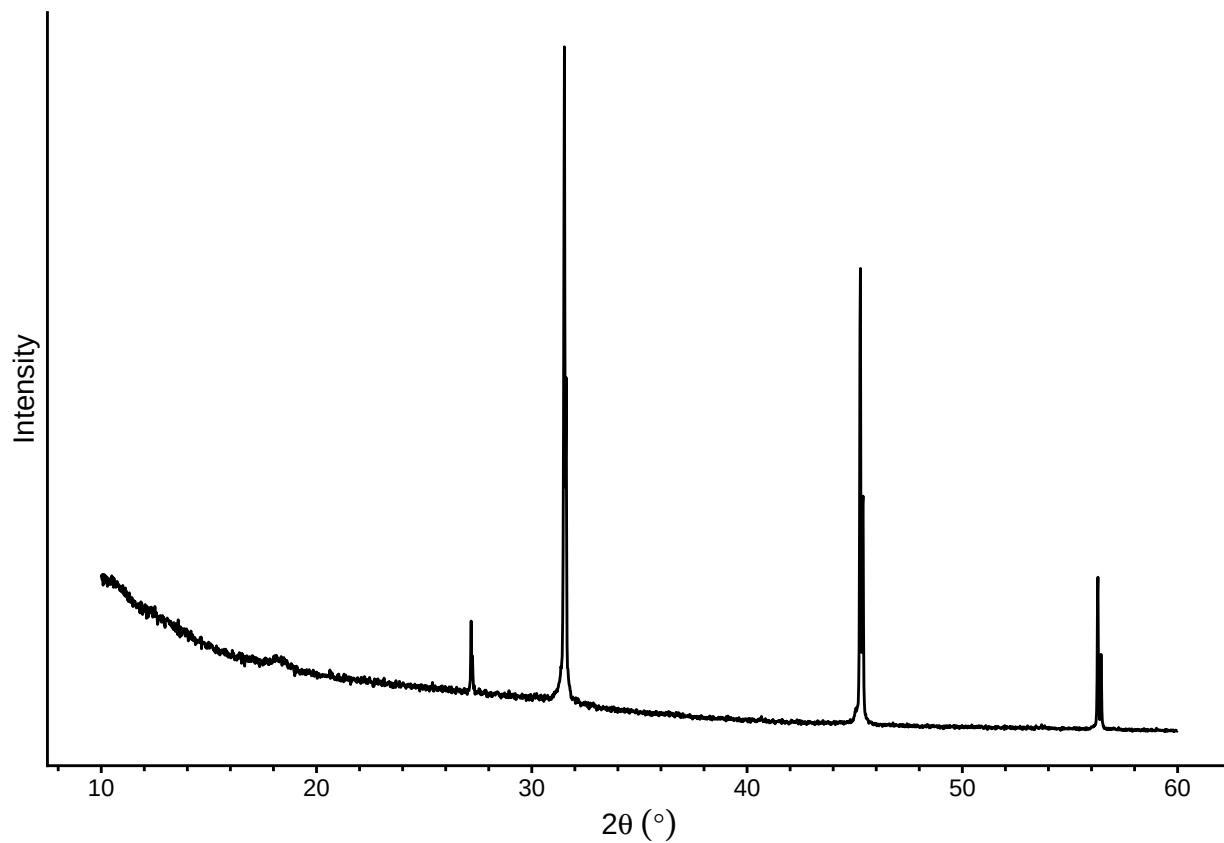
5. Overlay the Individual Points Over Your Box Plot From 3. and Adjust the Point Size and Transparency as Needed (3pts)

Description and Interpretation: I overlaid individual data points on the box plot containing dry weights of plants under different treatment conditions. The default point size and transparency provided enough contrast over the box plot outlines, and therefore did not need to be changed.



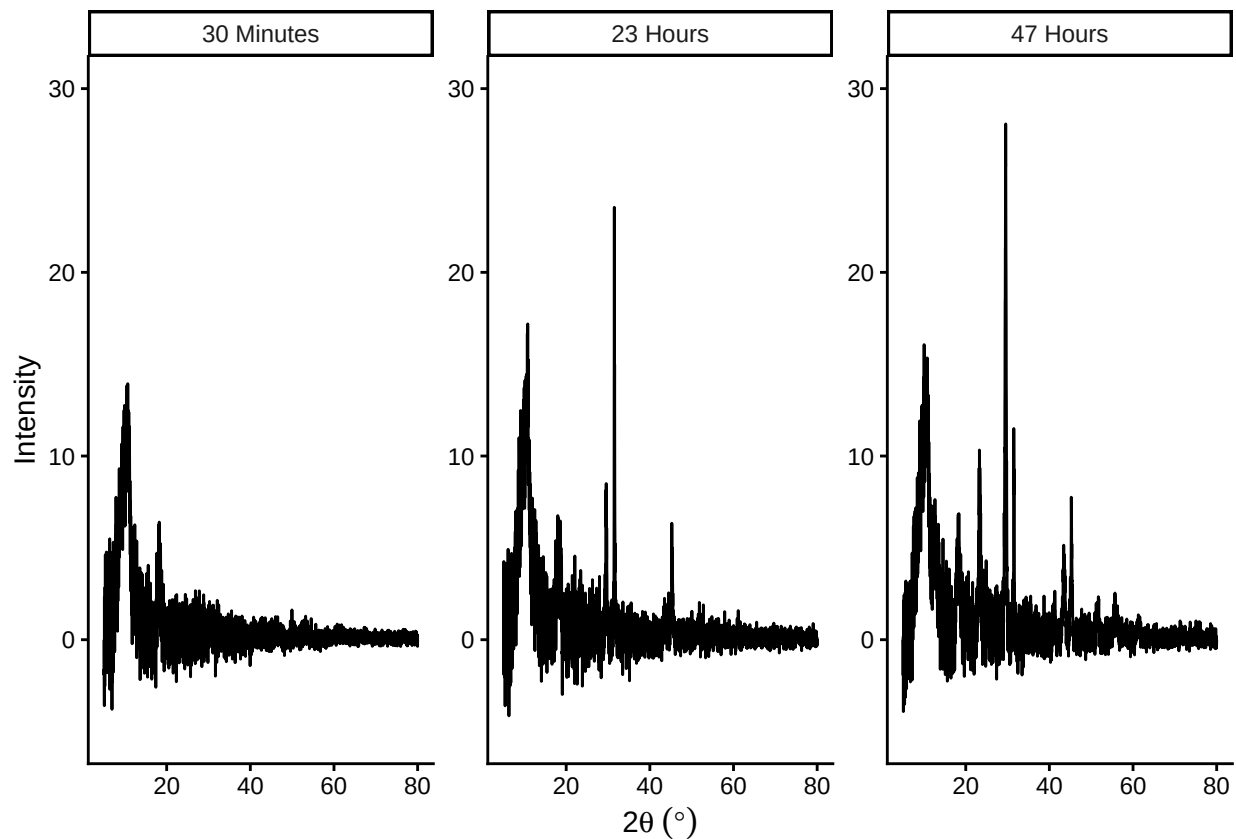
6. Make a plot and modify the scale in a beneficial way using `scale_x_continuous()`, `scale_y_continuous()`, and `coordinate_cartesian()`. Explain how your modifications to the scales improved the plot. (4pts)

Description and Interpretation: I loaded powder x-ray diffraction (PXRD) data that I collected. When plotting PXRD (intensity versus angle of detection) data for publication, it is standard to have labeled major ticks every 10 degrees and unlabeled minor ticks every 2 degrees. It is also standard to have no ticks on the y-axis, as it is unitless “intensity.”



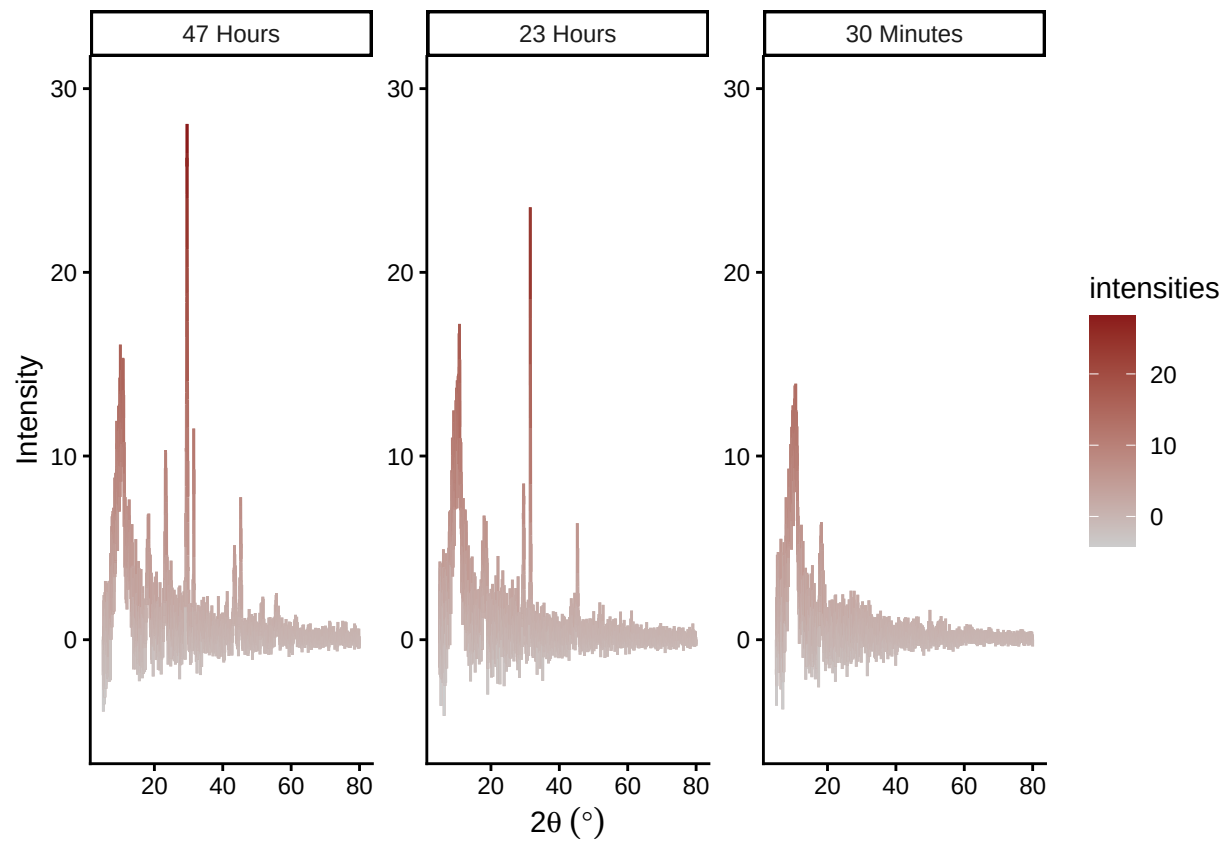
7. Make a New (don't use any plots from 1-4) Plot with `facet_wrap()` (4pts)

Description and Interpretation: This is also PXRD data collected at different timepoints (30 min, 23 hr, and 47 hr). I pivoted the data to create a new variable of time since reaction started, then used `facet_wrap()` to plot the pxrd spectra from each timepoint side-by-side.



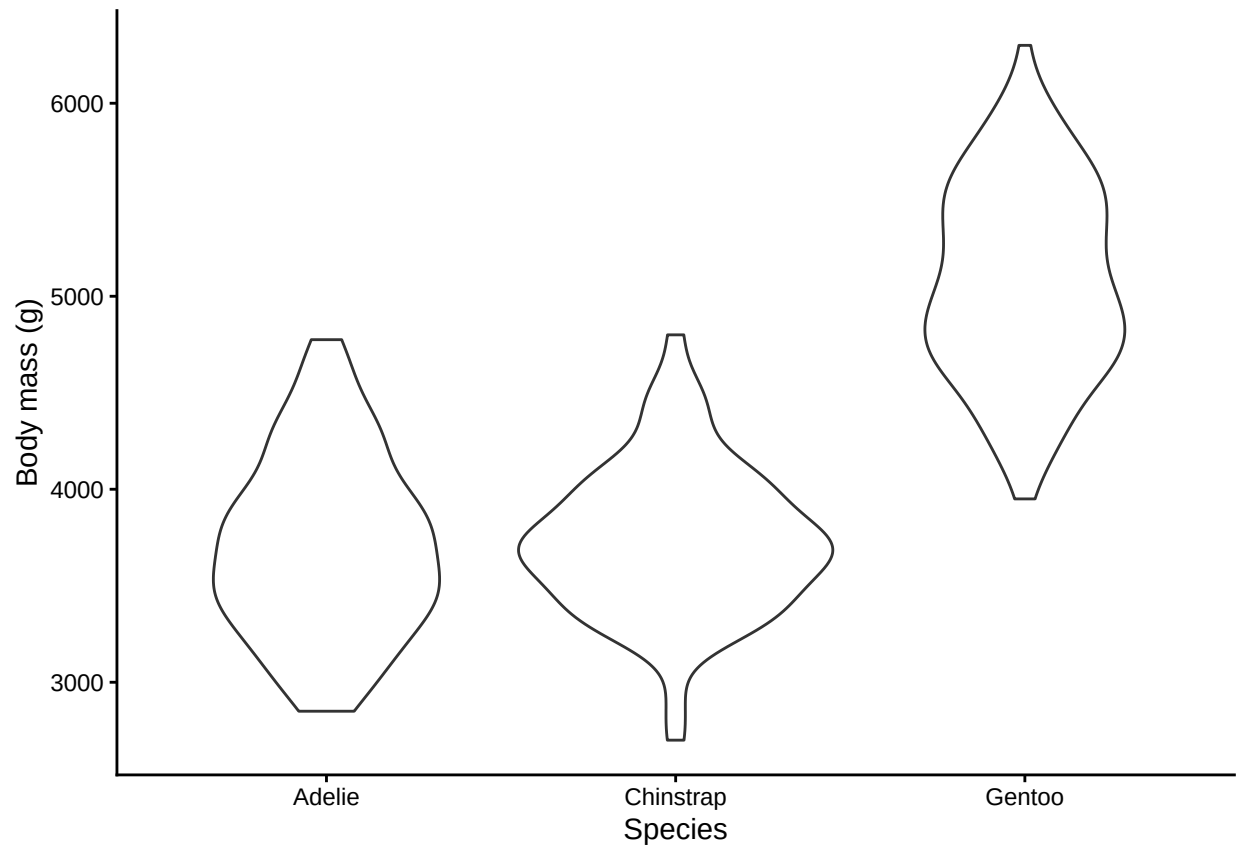
8. Using the plot from 7, scale the colors and reorder your facet to reverse order. (4pts)

Description and Interpretation: I reversed the order of my PXRD spectra so that they are listed in descending rather than ascending time increments. It makes no sense to do this, but that's what the question asked me to do. I also included a color gradient so that higher intensity values become more red.



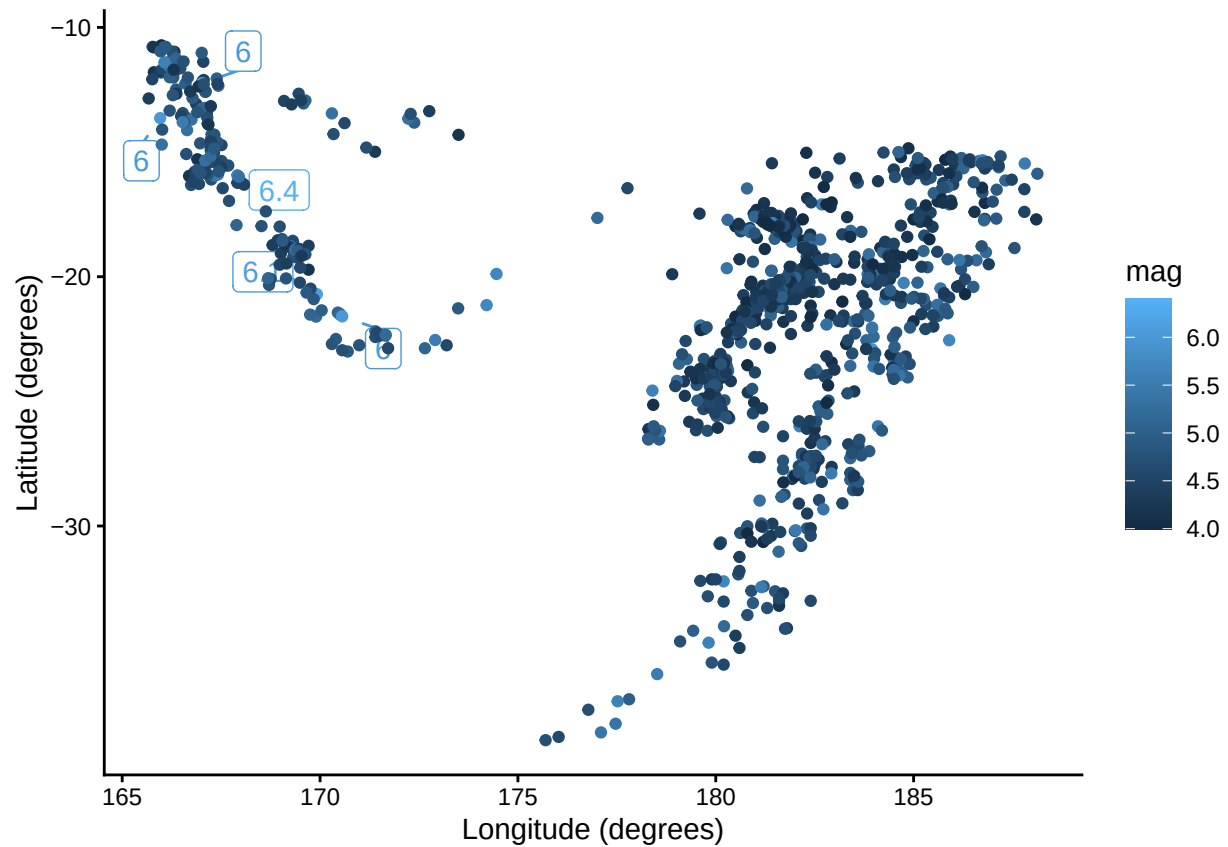
9. Make a Plot Using a Geometry That Was Not Covered in Class (4pts)

Description and Interpretation: I loaded a dataset containing the body mass in grams of penguins from different species that were found in the Palmer Archipelago. I plotted their body masses versus species in a violin plot, which shows the shape and distribution of the data.



10. Make a Plot Using a ggplot Extension That Was Not Covered in Class. Explain what the extension does. (4pts)

Description and Interpretation: I loaded a dataset containing latitude versus longitude values of earthquakes colorcoded by their magnitudes. I used the package `ggrepel`, which should repel overlapping text labels. However, I found that I still had issues with text labels overlapping with datapoints, despite extensive attempts to optimize `point.padding`.



Appendix

```
knitr::opts_chunk$set(echo = F, message = F, warning = F)
####
#00#
####
library(knitr)
library(tidyverse)
library(ggplot2)
library(readxl)
library(ggpubr)

####
#01#
####
BOD %>%
  ggplot(aes(x = Time, y = demand)) +
  geom_point(size = 3, alpha = 0.5) +
  labs(x = "Time (days)", y = "Oxygen demand (mg/l)")
####
```

```

#02#
####
BOD %>%
  ggplot(aes(x = Time, y = demand)) +
  geom_point(size = 3, alpha = 0.5) +
  geom_smooth(method = "lm", color = "black") +
  labs(x = "Time (days)", y = "Oxygen demand (mg/l)") +
  stat_regline_equation(label.x = 6, label.y = 22)

####
#03#
####
BOD %>%
  ggplot(aes(x = Time, y = demand, colour = demand)) +
  geom_point(size = 3,
             alpha = 1,
             shape = 18) +
  geom_smooth(method = "lm", color = "black") +
  labs(x = "Time (days)", y = "Oxygen demand (mg/l)") +
  stat_regline_equation(label.x = 6, label.y = 22) +
  scale_color_gradient(low = "firebrick1",
                      high = "firebrick4")

####
#04#
####
PlantGrowth %>%
  ggplot(aes(x = group, y = weight)) +
  geom_boxplot(width = 0.8)

####
#05#
####
PlantGrowth %>%
  ggplot(aes(x = group, y = weight)) +
  geom_boxplot(width = 0.8) +
  geom_jitter()

####
#06#
####
pxrd_7day_rxn = read_excel("20260407_no_peptide_7day_rxn_exported.xlsx")

names(pxrd_7day_rxn)[1:2] = c("2theta", "intensity")

pxrd_7day_rxn %>%
  ggplot(aes(x = `2theta`, y = `intensity`)) +
  geom_line() +
  scale_x_continuous(breaks = seq(0, 60, 10), minor_breaks = seq(0, 60, 2), guide = guide_axis(minor.ticks = FALSE)) +
  scale_y_continuous(labels = NULL) +
  labs(x = expression(2*theta~(degree)), y = "Intensity") +
  theme_classic() +
  theme(axis.ticks.y = element_blank())

```

```

#07#
####
pxrd_timecourse = read_excel("no_peptide_timecourse.xlsx")

pxrd_timecourse_long = pxrd_timecourse %>%
  pivot_longer(cols = c(`30`, `1380`, `2820`),
    names_to = "time_minutes",
    values_to = "intensities")

pxrd_timecourse_long$time_minutes = factor(pxrd_timecourse_long$time_minutes,
  levels = c("30", "1380", "2820"))
new_labels = c("30" = "30 Minutes", "1380" = "23 Hours", "2820" = "47 Hours")

pxrd_timecourse_long %>%
  ggplot(aes(x = `2theta`, y = intensities)) +
  facet_wrap(~ time_minutes,
    labeller = as_labeller(new_labels),
    scales = "free_y") +
  scale_y_continuous(limits = c(-5, 30)) +
  labs(x = expression(2*theta~(degree)), y = "Intensity") +
  geom_line() +
  theme_classic()

#08#
####
pxrd_timecourse_long$time_minutes = factor(pxrd_timecourse_long$time_minutes,
  levels = c("2820", "1380", "30"))
new_labels = c("30" = "30 Minutes", "1380" = "23 Hours", "2820" = "47 Hours")

pxrd_timecourse_long %>%
  ggplot(aes(x = `2theta`, y = intensities, colour = intensities)) +
  facet_wrap(~ time_minutes,
    labeller = as_labeller(new_labels),
    scales = "free_y") +
  scale_y_continuous(limits = c(-5, 30)) +
  scale_color_gradient(low = "grey80", high = "firebrick4") +
  labs(x = expression(2*theta~(degree)), y = "Intensity") +
  geom_line() +
  theme_classic()

####
#09#
####
penguins %>%
  ggplot(aes(x = species, y = body_mass)) +
  labs(x = "Species", y = "Body mass (g)") +
  geom_violin() +
  theme_classic()

####
#10#
####

```

```

library(ggrepel)

quakes %>%
  ggplot(aes(x = long, y = lat, colour = mag)) +
  geom_label_repel(
    data = ~ dplyr::filter(.x, mag >= 6),
    aes(label = mag),
    force = 5,
    box.padding = 0.5,
    point.padding = 0.5,
    min.segment.length = 0,
    max.overlaps = Inf) +
  labs(x = "Longitude (degrees)", y = "Latitude (degrees)") +
  geom_point() +
  theme_classic()

```